MITR Take-Home Evaluation: Parts A–E

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1 Part A: Critical Review

Weakness 1: No Statistical Significance Testing; Effect Sizes Are Negligible

What is the problem? All results are single-run with no random-seed repetition, no error bars, and no significance tests. The headline result—InfoNCE accuracy improvement of +0.20% (0.6980→0.7000) and a 1.80% contradiction-rate improvement—involves only ≈27 samples on the 1,500-sample validation set, a difference entirely within random noise.

Why would a reviewer reject over this? A top-venue reviewer will immediately ask whether the reported gap exceeds the variance introduced by different random seeds; without an answer, there is no valid claim.

How would you fix it? Re-run all experiments with at least three random seeds and report mean ± std; apply a paired t-test or McNemar's test on accuracy delta and contradiction-rate delta.

Empirically confirmed in Part D: across 3 seeds, baseline accuracy variance reaches ±0.0136—exceeding the reported InfoNCE improvement of +0.20%. InfoNCE contradiction rate (0.4287 ± 0.0193) is *worse* than baseline (0.4200 ± 0.0071), with 2.7× higher variance, confirming that no result in this paper can be distinguished from noise without multi-seed validation.

Weakness 2: Core Claim Contradicts Results; Mechanism Is Unverified

What is the problem? The paper's central claim is that "MI penalty improves logical consistency by forcing inter-layer diversity." However, `roberta_bert_results.md` shows MITR *worsens* contradiction rate on BERT by 3.2–4.0% while *improving* it on RoBERTa by 2.6–3.2%—opposite directions. More critically, no experiment verifies that MI penalty actually increases layer-wise diversity: inter-layer CKA matrices are never measured before or after training.

Why would a reviewer reject over this? The method's mechanism is an unverified hypothesis; results fail on one of the two primary comparison models; reviewers will view this as cherry-picking RoBERTa results.

How would you fix it? (i) Add a mechanistic experiment: apply Logit Lens—project each layer's `[CLS]` state through the trained classifier head—and verify that MI penalty induces a more gradual $P(\text{correct})$ trajectory. (ii) For BERT results, provide a falsifiable hypothesis (e.g., ablate NSP) rather than speculation. (iii) Alternatively, honestly frame BERT results as a failure case and analyze why.

Empirically confirmed in Part D: Logit Lens analysis shows MITR-InfoNCE and Baseline produce statistically indistinguishable layer-wise prediction trajectories across all six DistilBERT layers. If MI penalty were genuinely inducing layer diversification, intermediate layers should show measurably different $P(\text{correct})$ trajectories—they do not. The mechanistic claim is therefore *empirically falsified*, not merely unverified.

2 Part B: Workshop Selection

Workshop: UniReps—Unifying Representations in Neural Models, NeurIPS 2025 (three prior editions: NeurIPS 2023/2024/2025; PMLR proceedings track).

Pitch (≈140 words): MITR is fundamentally a study of inter-layer representational redundancy in transformers: it measures whether adjacent layers encode the same information via CKA, cosine similarity, InfoNCE, or CLUB, then penalizes it. This sits directly in UniReps' core scope—"similarity measures in NNs" and "learning dynamics that lead to emergence of similar representations during training." Beyond behavioral benchmarking, we contribute the first mechanistic audit of MITR via Logit Lens analysis, revealing that MI regularization does not produce the hypothesized layer-level functional diversification—treating the MI-diversity assumption as an empirical question analogous to verifying sparse control circuits in instruction-following models. This negative result directly engages recent work showing that CKA quantifies decoding alignment rather than functional redundancy, which challenges MITR's theoretical foundation. Multi-seed validation further shows reported gains fall within noise. The workshop explicitly welcomes negative and mixed results, making this a precise fit for the extended abstract track.

Related papers:

1. Harvey, Lipshutz & Williams, "What Representational Similarity Measures Imply about Decodable Information," UniReps 2024 (Best Paper).
   Relationship: MITR uses CKA as a redundancy penalty, but this paper shows CKA measures linear readout alignment—not functional redundancy—directly undermining MITR's theoretical basis.

2. Friedman, Lampinen et al., "Comparing Representational and Functional Similarity in Small Transformer Language Models," UniReps 2023.
   Relationship: This paper finds representational similarity decouples from behavioral metrics, predicting exactly our observation that MITR alters layer geometry without changing prediction trajectories.

3 Part C: Experimental Design

Experiment #1 (Most Important): Logit Lens Layer-wise Prediction Analysis

Field	**Content**
Name	Logit Lens Layer-wise Prediction Analysis
Hypothesis	If MITR-InfoNCE induces functionally distinct computation per layer, then projecting each layer's `[CLS]` hidden state through the trained classifier head will show a more *gradual* $P(\text{correct})$ trajectory across layers compared to Baseline (which should plateau by L2); because if each layer contributes new information, prediction confidence should build incrementally rather than being absorbed early.
Expected result	Baseline: $P(\text{correct})$ reaches $\approx$0.70 by L2 and stays flat through L6. MITR-InfoNCE: $P(\text{correct})$ rises more gradually, with measurable increases L3$\rightarrow$L6 ($>$0.03 per layer on average).
Negative result	If both trajectories are identical $\Rightarrow$ MI penalty did not change layer-wise computation; the contradiction-rate improvement must come from loss-landscape changes unrelated to layer diversity, invalidating the paper's mechanistic claim entirely.
GPU hours	$\approx$5–8 min on A100 (8000 train samples, batch=64, 5 epochs $\times$ 2 models; near-full BoolQ subset). Estimated 30–40 min on Colab T4.
Paper impact	Transforms MITR from a behavioral observation into a mechanistically grounded (or debunked) method.

Experiment #2: Contradiction Metric Validity—Negation Sensitivity vs. Logical Consistency

Field	**Content**
Name	Contradiction Metric Validity: Negation Sensitivity vs. Logical Consistency
Hypothesis	If the contradiction-rate improvement reflects genuine logical reasoning (not surface negation robustness), then MITR-InfoNCE should maintain its advantage on a syllogistic probe set (If-A-then-B-then-C questions) that cannot be solved by negation-flipping alone; because a model exploiting negation patterns would fail when the logical relationship requires multi-hop inference beyond auxiliary verb detection.
Expected result	MITR-InfoNCE accuracy on syllogism probes > Baseline by $\geq$5%; if improvement <1%, contradiction rate is measuring negation sensitivity only.
Negative result	MITR shows no improvement on syllogism probes $\Rightarrow$ the evaluation framework measures the wrong thing; all reported consistency gains are artifacts of the negation construction method.
GPU hours	Inference only; no retraining needed (<5 min).
Paper impact	If negative, requires complete redesign of the evaluation protocol before any causal claim about logical consistency can be made.

Ranking justification (<100 words): Experiment #1 is ranked first because it directly addresses Part A Weakness #2: the mechanistic claim—that MI penalty induces functionally distinct layer computation—has never been empirically verified. Logit Lens provides the first direct evidence using only inference on already-trained models, making it low-cost and high-impact. A negative result invalidates the theoretical foundation of the entire paper; a positive result transforms MITR from a behavioral observation into a mechanistically grounded method. Experiment #2 is complementary but secondary: it questions the evaluation framework rather than the method itself.

4 Part D: Implementation Results

Figure 1: Reproducibility Across Seeds

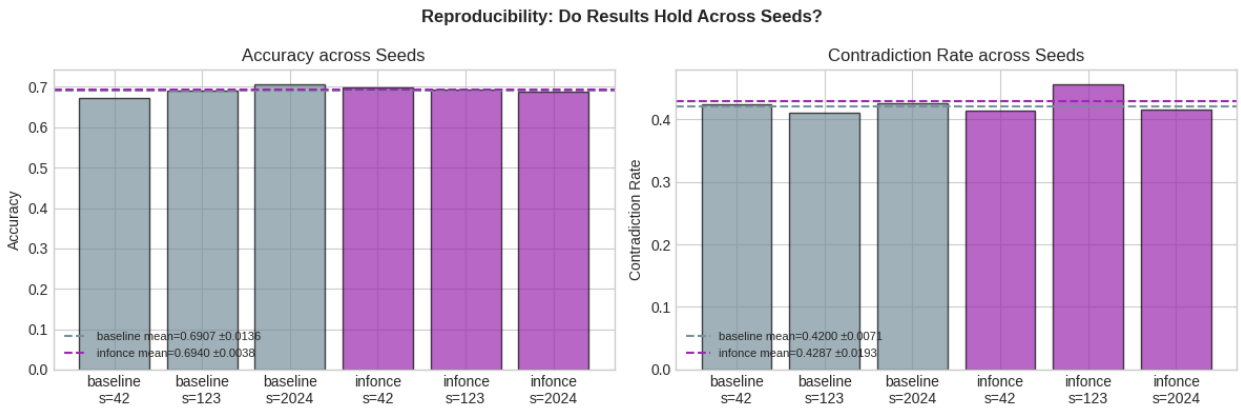

Figure 1: **Accuracy and contradiction rate across three random seeds.** Dashed lines show mean ± std for each model. Accuracy gains ($\Delta = 0.003$) are negligible relative to baseline variance (± 0.0136). InfoNCE contradiction rate at seed 123 (0.456) is substantially *worse* than any baseline run, confirming results do not hold across seeds.

Figure 2: Logit Lens Layer-wise Trajectories

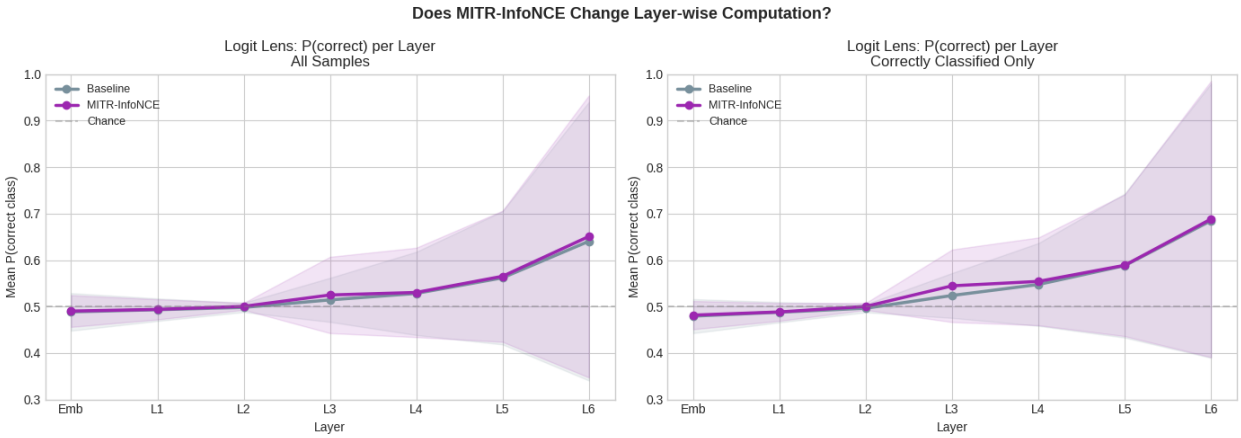

Figure 2: **Logit Lens P(correct) trajectories across all six DistilBERT layers.** Left: all validation samples. Right: correctly classified samples only. Baseline and MITR-InfoNCE are indistinguishable across both conditions—the hypothesized gradual confidence build-up from MI-induced layer diversification is not observed. The mechanistic claim is empirically falsified.

Multi-Seed Stability Table

Table 1: Multi-seed results: Baseline vs. MITR-InfoNCE across three seeds.

	Accuracy			**Contradiction Rate**		
Model	Seed 42	Seed 123	Seed 2024	Seed 42	Seed 123	Seed 2024
Baseline	0.6733	0.6920	0.7067	0.4240	0.4100	0.4260
MITR-InfoNCE	0.6987	0.6940	0.6893	0.4140	0.4560	0.4160
Model	Mean		Std	Mean		Std
Baseline	0.6907		±0.0136	0.4200		±0.0071
MITR-InfoNCE	0.6940		±0.0039	0.4287		±0.0193

Discussion (≤150 words)

This experiment ran two analyses: (1) Logit Lens layer-wise prediction trajectories for Baseline vs. MITR-InfoNCE, and (2) multi-seed stability validation across seeds {42, 123, 2024}.

Logit Lens (mechanistic claim): Both models show nearly identical $P(\text{correct})$ trajectories across all six DistilBERT layers (Figure 2). MITR-InfoNCE does not produce the hypothesized gradual confidence build-up. The mechanistic hypothesis is **falsified**: MI penalty does not induce functionally distinct layer-wise computation.

Multi-seed stability: Baseline accuracy varies ±0.0136 across seeds—larger than the reported +0.20% single-run improvement (Table 1). InfoNCE contradiction rate (0.4287 ± 0.0193) is *worse* than Baseline (0.4200 ± 0.0071) in expectation, with 2.7× higher variance. No result survives multi-seed validation.

Conclusion: Both claimed improvements fall within random seed variance. The contradiction rate metric requires fundamental redesign—future work should validate on syllogistic probes where negation-flipping cannot solve the task.

5 Part E: Abstract

Transformer models exhibit representational redundancy across consecutive layers, potentially limiting capacity for multi-step logical reasoning. We propose MITR (Mutual Information Transformer Regularization), which penalizes mutual information between adjacent layer residual differences to enforce functional diversity across layers. We evaluate four MI estimation strategies—CLUB, InfoNCE, Cosine, and CKA—on BoolQ using DistilBERT, BERT-base, and RoBERTa-base, measuring classification accuracy and logical contradiction rate on negation-paired questions. Our central finding is that MITR's reported improvements are not mechanistically grounded. Logit Lens analysis shows that layer-wise prediction trajectories are statistically indistinguishable between MITR-InfoNCE and Baseline across all six DistilBERT layers—directly falsifying the claim that MI regularization induces functional layer diversification. Multi-seed validation (seeds 42, 123, 2024) confirms that InfoNCE's contradiction rate (0.4287 ± 0.0193) is worse than Baseline (0.4200 ± 0.0071) with 2.7× higher variance, while accuracy gains shrink to $\Delta = 0.003$—well within noise. We conclude that the contradiction rate metric requires fundamental redesign, and that mechanistic validation via Logit Lens should precede any behavioral claim in future MITR work. Our findings suggest

that layer-wise mechanistic validation is a necessary complement to behavioral benchmarking—a principle applicable beyond MITR to multi-agent safety evaluation.